

Nurture Course COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-NURTURE 2023

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	3	4	1	1	2	3	3	4	3	3	3	3	4	3	4	4	4	2	3	3	2	3	3	2	1	2	4	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	4	4	1	2	2	3	4	2	1	3	1	3	4	4	2															



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HINT - SHEET

1. **Ans (3)**

$$\frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$$

$$\frac{W}{6} = \frac{400 - 300}{400}$$

$$\Rightarrow W = \frac{6}{4} = 1.5 \text{ kCal}$$

2. **Ans (3)**

$$K_{eq} = \frac{K_1 + K_2}{2} = \frac{K + 2K}{2} = \frac{3K}{2}$$

3. **Ans (4)**

$$PV = \mu RT \quad \text{where } \mu = \frac{5}{28} \text{ moles}$$

4. **Ans (1)**

$$\Delta U = \mu C_V \Delta T \quad \& \quad 0 = W + \Delta U$$

$$\Rightarrow \Delta U = -6R \quad (\because W = 6R)$$

$$\text{Therefore } -6R = 1 \left(\frac{R}{\gamma - 1} \right) \Delta T = \frac{5}{2} R \Delta T$$

$$\Rightarrow \Delta T = -2.4 \Rightarrow T_{\text{final}} = (T - 2.4)K$$

5. **Ans (1)**

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \frac{6 \times 10^4}{Q_2} = \frac{500}{400} \Rightarrow Q = 4.8 \times 10^4$$

6. **Ans (2)**

$$\lambda_{m_1} T_1 = \lambda_{m_2} T_2$$

$$7000 \times 2500 = \lambda_{m_2} \times 3500$$

$$\Rightarrow \lambda_{m_2} = 5000 \text{ Å}$$

7. **Ans (3)**

degree of freedom $f = 5$ for diatomic gas
 So, $C_V = \frac{5}{2}R$ and $C_P = \frac{5}{2}R + R = \frac{7R}{2}$

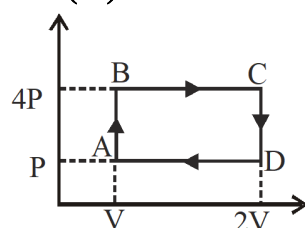
8. **Ans (3)**

$$C_P = \frac{5}{2}R$$

$$\Rightarrow C_V = C_P - R = \frac{3}{2}R$$

$$\Rightarrow \gamma = \frac{C_P}{C_V} = \frac{5}{3}$$

9. **Ans (4)**



in cyclic process,
 $W = \text{area enclosed by curve}$
 $= 3P \times V = 3PV$

10. **Ans (3)**

$$\frac{1}{3} = \frac{T_1 - T_2}{T_1}$$

$$\frac{2}{3} = \frac{T_1 - (T_2 - 62)}{T_1} = \frac{T_1 - T_2}{T_1} + \frac{62}{T_1}$$

$$\frac{2}{3} = \frac{1}{3} + \frac{62}{T_1}, \quad \frac{1}{3} = \frac{62}{T_1},$$

$$T_1 = 186 \text{ K}$$

$$T_2 = 124 \text{ K}$$

11. **Ans (3)**

$$\frac{x - MP}{BP - MP} = \frac{^{\circ}C}{100}$$

$$\Rightarrow \frac{x - 39}{139 - 39} = \frac{39}{100}$$

$$\Rightarrow x = 78^{\circ}W$$

12. **Ans (3)**

$$\frac{\rho}{P} = \frac{M_w}{R \times 293} = x$$

$$\Rightarrow \frac{M_w}{R} = 293x$$

$$\frac{\rho'}{P'} = \frac{M_w}{R \times 393} = \frac{293}{393}x$$

13. **Ans (3)**

$$E \propto T^4$$

$$E_2 = E_1 \left(\frac{T_2}{T_1} \right)^4$$

$$E_2 = 2 \left(\frac{1200}{400} \right)^4 = 162 \text{ units}$$

14. **Ans (4)**

$$\text{FLOT} \quad Q = W + \Delta U$$

15. **Ans (3)**

During melting, Volume = constant

$$A_1 \ell_1 = A_2 \ell_2$$

$$\ell_2 = \frac{(\pi r^2) \times \ell}{\pi \left(\frac{r}{3} \right)^2} = 9\ell \text{ and } \frac{Q}{t} \propto \frac{A}{\ell}$$

$$\text{Hence } \frac{Q'}{Q} = \frac{\pi \left(\frac{r}{3} \right)^2 \times \ell}{\pi (r^2) \times 9\ell} = \frac{1}{81}$$

$$\Rightarrow Q' = \frac{Q}{81}$$

16. **Ans (4)**

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow P_2 = \left(\frac{V_1}{V_2} \right)^\gamma \times P_1$$

$$P_2 = \left(\frac{V}{V/27} \right)^{5/3} \times P_1$$

$$\Rightarrow P_2 = 243P_1$$

17. **Ans (4)**

$$W = +250 \text{ J} \quad T = \text{constant} (\Delta U = 0)$$

According to FLOT

$$Q = W + \Delta U \Rightarrow Q = W = +250 \text{ J}$$

It means 250 J of heat has been added to the gas.

18. **Ans (4)**

As from adiabatic process

$$P^{1-\gamma} T^\gamma = \text{constant}$$

$$\left(\frac{P_2}{P_1} \right)^{1-\gamma} = \left(\frac{T_1}{T_2} \right)^\gamma = \left(\frac{200}{800} \right)^\gamma$$

$$\left(\frac{P_2}{P_1} \right)^{1-7/5} = \left(\frac{200}{800} \right)^{7/5}$$

$$\Rightarrow \left(\frac{P_1}{P_2} \right)^2 = \left(\frac{1}{4} \right)^7$$

$$P_2 = 2^7 \times 2 = 2^8 = 256 \text{ atm.}$$

19. **Ans (2)**

$$\Delta S = \frac{Q}{T} = \frac{mL_f}{T} = \frac{2000 \times 80}{273} = 586 \text{ cal/K}$$

20. **Ans (3)**

$$E = \frac{Q'}{At} = \sigma T^4$$

Rate of energy production is $Q \left[Q = \frac{Q'}{t} \right]$

$$\frac{Q}{A} = \sigma T^4 \Rightarrow T = \left(\frac{Q}{4\pi(R/2)^2 \sigma} \right)^{1/4}$$

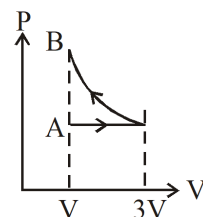
$$= \left(\frac{Q}{\pi R^2 \sigma} \right)^{1/4}$$

21. **Ans (3)**

I $\rightarrow$ Isobaric expansion from $V \rightarrow 3V$ ($P = \text{constant}$)

II $\rightarrow$ compressed at constant temperature from

$3V \rightarrow V$



22. **Ans (2)**

In cyclic process ($\Delta U = 0$)

According to FLOT

$$Q = W$$

$W = \text{Area enclosed}$

$Q = -PV$ ($-ve$ mean heat rejected)

So answer is PV

23. **Ans (3)**

Heating curve.

24. **Ans (3)**

$$\frac{dQ}{dt} = \frac{KA(T_1 - T_2)}{L}$$

$$dQ = \frac{KA(T_1 - T_2)dt}{L} = mL$$

$$\frac{K \times 0.28 \times 100 \times 60 \times 60}{0.1} = 3.72 \times 3.36 \times 10^5$$

$$K = 1.24 \text{ J/m/s/}^\circ\text{C}$$

25. **Ans (2)**

Number of moles in 1g He = $\frac{1}{4}$

Amount of heat energy required to raise its temperature from T_1K to T_2K

$$= \mu C_p \Delta T$$

$$= \left(\frac{1}{4}\right) \left(\frac{5}{2}R\right) (T_2 - T_1)$$

$$= \frac{5}{8} k_B N_A (T_2 - T_1)$$

26. **Ans (1)**

$$C_p - C_v = R \text{ and } g = \frac{C_p}{C_v} \Rightarrow C_p = \frac{\gamma R}{\gamma - 1}$$

27. **Ans (2)**

According to question,

$$P \propto T^4$$

$$P = aT^4 \quad a = \text{constant}$$

$$PT^{-4} = a \quad \dots(1)$$

Relation between P and T in adiabatic process

$$PT^{\frac{\gamma}{1-\gamma}} = \text{constant} \quad \dots(2)$$

By comparing (1) and (2)

$$\frac{\gamma}{(1-\gamma)} = -4$$

$$\Rightarrow \gamma = \frac{4}{3} \quad \text{i.e.} \quad \frac{C_p}{C_v} = \frac{4}{3}$$

28. **Ans (4)**

$W \rightarrow$ In cyclic process.

$$W = \frac{1}{2} (7-2) \times 10^{-3} \times (5-2) \times 10^5 = 750 \text{ J}$$

Cycle is clockwise $W = +ve$

29. **Ans (3)**

$$m_1 L_v + m_1 s \Delta q = m_2 s \Delta q$$

$$m \times 540 + m \times 1 \times 20 = 40 \times 1 \times 70$$

$$m_1 \times 560 = 2800$$

$$m_1 = 5 \text{ g}$$

$$\therefore \text{Total mass of water} = 40 + 5 = 45 \text{ g}$$

30. **Ans (1)**

$$\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_0 \right)$$

$T_0 \rightarrow$ surrounding temperature.

31. **Ans (4)**

In isothermal process : $P_1 V_1 = P_2 V_2$

$$PV = P_2(4V) \Rightarrow P_2 = \frac{P}{4}$$

In adiabatic process

$$\left(\frac{P}{4}\right) (4V)^\gamma = P' (32V)^\gamma$$

$$\Rightarrow P' = \frac{P}{4} \times \frac{1}{32} = \frac{P}{128}$$

32. **Ans (4)**

$$W = W_{AEDA} + W_{CEBC} = 0$$

OR

$$W = \frac{1}{2} [5P_0 + P_0] \times V_0 - 5P_0 V_0 + \frac{1}{2} [5P_0 + P_0] V_0 - P_0 V_0$$

$$= 0$$

33. **Ans (1)**

$$\Delta U = \mu C_v \Delta T \quad \& \quad T = \frac{PV}{\mu R}$$

$$\text{so } \Delta T = T_2 - T_1 = \frac{P_2 V_2 - P_1 V_1}{\mu R}$$

$$\text{so } \Delta U = \frac{\mu R}{\gamma - 1} \left(\frac{P_2 V_2 - P_1 V_1}{\mu R} \right) = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$\Rightarrow \Delta U = \frac{-8 \times 10^3}{2/5} = -20 \text{ kJ}$$

34. **Ans (2)**

For Engine & refrigerators operating between two same temperatures

$$\eta = \frac{1}{1 + \beta}$$

$$\Rightarrow \frac{1}{11} = \frac{1}{1 + \beta} \Rightarrow \beta = 10$$

$$10 = \frac{Q_2}{9} \text{ (From the principle of refrigerator)}$$

$$9 = \frac{Q_2}{10} \Rightarrow Q_2 = 90 \text{ J}$$

35. **Ans (2)**

In cyclic process ABCA,

$$\Delta U_{\text{cyclic}} = 0$$

$$Q_{\text{cyclic}} = W_{\text{cyclic}}$$

$$Q_{AB} + Q_{BC} + Q_{CA} = \text{area of closed loop}$$

$$400 + 100 + Q_{CA} = \frac{1}{2} \times (2 \times 10^{-3}) \times 4 \times 10^4$$

$$400 + 100 - Q_{AC} = 40$$

$$Q_{AC} = 460 \text{ J}$$

36. **Ans (3)**

$$\frac{dQ}{dt} \propto (T_1 - T_2) \quad (T_1 > T_2)$$

Here temperature difference in both case is same (i.e. 10°C)

So, rate of heat flow will also be same

$$\text{So, } \frac{dQ}{dt} = 4 \text{ J/s}$$

37. **Ans (4)**

According to ideal gas equation

$$P = \frac{\rho RT}{M_w}$$

$$\Rightarrow M_w = \frac{\rho RT}{P}$$

$$\text{so } \frac{M_A}{M_B} = \frac{\rho_A}{\rho_B} \cdot \frac{T_A}{T_B} \cdot \frac{P_B}{P_A} = (1/1.5) (1) \left(\frac{1}{2}\right)$$

$$\Rightarrow \frac{M_A}{M_B} = \frac{1}{3}$$

38. **Ans (2)**

Molecular mass $M_w = 4.0 \text{ g}$

$$v_{\text{sound}} = \sqrt{\frac{\gamma RT}{M_w}}$$

$$\Rightarrow \gamma = \frac{M_w v^2}{RT} = 1.6$$

So, $C_p = \gamma C_v$

$$\Rightarrow 8 = 1.6 C_v = 5 \text{ J K}^{-1} \text{ mol}^{-1}$$

39. **Ans (1)**

Coefficient of performance of refrigerator

$$\text{COP} = \frac{T_L}{T_H - T_L}$$

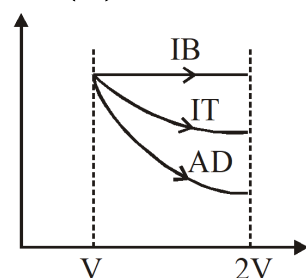
Where $T_L \rightarrow$ lower Temperature

& $T_H \rightarrow$ Higher Temperature

$$\text{So, } 6 = \frac{T_L}{T_H - T_L}$$

$$\Rightarrow T_H = \frac{7}{6} T_L = \frac{7}{6} \times 252 = 284 \text{ K} = 21^\circ\text{C}$$

40. **Ans (3)**



work done on the gas

$$W_{\text{isochoric}} = 0$$

$$\text{and } W_{\text{adiabatic}} < W_{\text{isothermal}} < W_{\text{isobaric}}$$

41. **Ans (1)**

$$d_f = \frac{d_i}{(1 + \gamma \Delta T)}$$

fractional change

$$= \frac{d_i - d_f}{d_i} = 1 - \frac{d_f}{d_i}$$

$$= 1 - (1 + \gamma \Delta T)^{-1}$$

$$= 1 - (1 - \gamma \Delta T) \quad \because (1+x)^n \approx 1 + nx$$

$$= \gamma \Delta T = 5 \times 10^{-4} \times 20^\circ\text{C} = 0.01$$

42. **Ans (3)**

$$\beta = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2} \quad (\text{Where } Q_2 \text{ is heat removed})$$

$$\Rightarrow \frac{600 \times 4.2}{W} = \frac{280}{308 - 280}$$

$$\Rightarrow W = 252 \text{ joule}$$

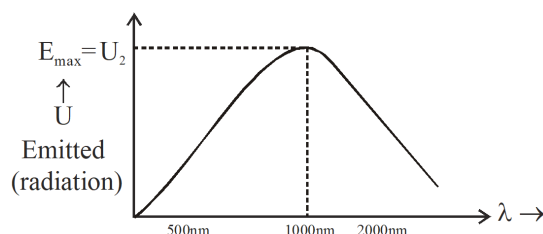
$$\Rightarrow \text{Power} = \frac{W}{t} = \frac{252 \text{ joule}}{1 \text{ sec}} = 252 \text{ watt.}$$

43. **Ans (4)**

Maximum amount of emitted radiation

corresponding to $\lambda_m = \frac{b}{T}$

$$\lambda_m = \frac{2.88 \times 10^6 \text{ nmK}}{2880 \text{ K}} = 1000 \text{ nm}$$



From the graph $U_2 > U_1$ and $U_2 > U_3$

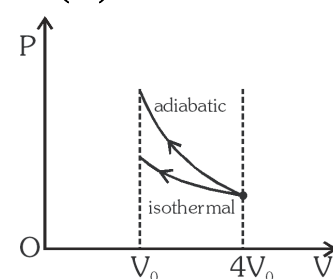
44. **Ans (4)**

$$v \propto \sqrt{T}$$

$$\Rightarrow \frac{v}{200} = \sqrt{\frac{500}{400}}$$

$$\Rightarrow v = 100\sqrt{5} \text{ m/s}$$

45. **Ans (2)**



W_{ext} = negative of area with volume-axis

$$W(\text{adiabatic}) > W(\text{isothermal})$$